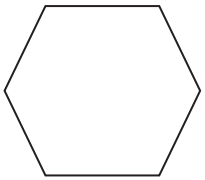
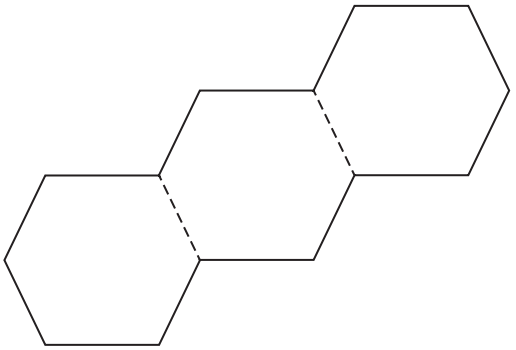
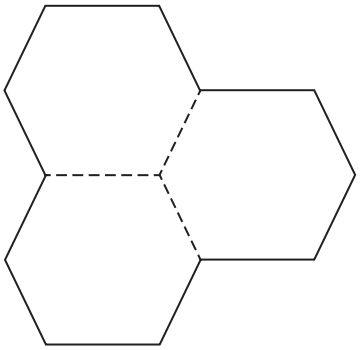
7 Here is a regular hexagon.



There are six identical hexagons.
Three of the hexagons are joined to make shape **A**.
The other three hexagons are joined to make shape **B**.



Shape **A**



Shape **B**

Which shape has the greater perimeter, shape **A** or shape **B**?
You must show how you get your answer.

Let one hexagon side have length s.

1 mark
Count exposed edges

Three separate hexagons have 18 sides.

A has 2 shared sides: perimeter = (18 – 4)s = 14s.

B has 3 shared sides: perimeter = (18 – 6)s = 12s.

Shape A has the greater perimeter, since 14s > 12s.

1 mark
A: 14 sides; B: 12

(Total for Question 7 is 2 marks)